

AlexNet Architecture

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Abstract

Today we are witnessing a deep learning revolution. New models are built and being tested faster than ever before. It all started at ImageNet 2012. A deep layered convolutional neural network won the contest, absolutely crushing the rest of the competition. The model is called AlexNet, and its unparalleled performance improvement of what came before is what changed the paradigm of deep learning forever. It completely proved that deep learning was practical.

Keywords: CNN, AlexNet, Normalization

1 Introduction

Until 2012, computer vision had a very slow progress due to the lack of funding and proof of practicality of deep learning.

In the context of convolutional neural networks, there is an deep neural network for image recognition and classification called AlexNet. It was made in the context of the ImageNet LSVRC-2012 contest and won that contest. It was made to improve machine learning methods to recognize objects in an image and prevent overfitting. It is similar to another CNN: LeNet-5, developed many years prior. The AlexNet was the result of training one of the largest convolutional neural networks to that date (2012) on the subsets of ImageNet and achieving one of the best results ever reported on those datasets (17% error) .

2 Background: the dataset

Until 2006, the world of computer vision was very underfunded, although there were a lot of researchers around the world focused on building better models. Progress was slow.

A young woman named Fei-Fei Li had just completed her PHD in Computer Vision at Caltech and started as a computer science professor at the University of Illinois. She noticed the focus on models and a lack of focus on data. So, she thought that the key to better model performance could be bigger datasets that reflected the diversity of the real world. She researched datasets and learned about a dataset called WordNet, which consisted of many English language terms organized into an ontological structure.

The ImageNet had over 12 million labeled high-resolution images belonging to 22000 categories. In the first annual competition ImageNet Large-Scale Visual Recognition Challenge, ILSVRC-2010, the images were all labeled, and there were 1.2 million images in the training set, 50000 validation images and 150000 in the test set.

ImageNet has variable-resolution images, so the AlexNet team had to rescale them to a fixed resolution, then trained the network on the centered RGB values of the pixels. No other pre-processing was made.

3 The architecture

The architecture of the network contains eight layers, five convolutional and three fully connected. AlexNet top-5 error is 17%, while the previous architecture LeNet-5 had a top-5 error of 26%.

3.1 Convolutional Layers

AlexNet consists of eight layers: five convolutional layers followed by three fully connected layers. The first convolutional layer has 96 kernels of size 11x11 with a stride of 4 pixels. Subsequent layers feature smaller kernels of 5x5 and 3x3. This architectural choice allowed AlexNet to capture various levels of detail from the input images.

3.2 Novel feature- ReLU activation function Nonlinearity

AlexNet popularized the use of Rectified Linear Units (ReLU) as activation functions. Unlike saturating nonlinearity functions like tanh and sigmoid, ReLUs do not saturate, leading to faster training and enabling the model to handle larger datasets. This choice was pivotal in the success of AlexNet, as demonstrated in experiments with the CIFAR-10 dataset.

ReLU stands for Rectified Linear Units. The comparisons highlight that ReLUs train several times faster than the saturating functions, enabling faster learning and making it feasible to experiment with larger neural networks and datasets, as demonstrated in experiments on CIFAR-10 dataset (picture above).

This is why ReLU functions were used for the training of the model.

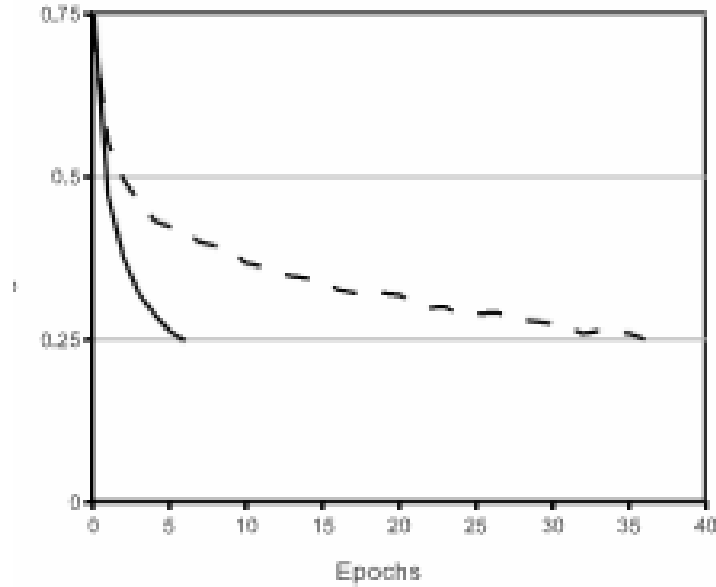


Fig. 1 ReLU() a atingir a perc de erro minima mais rápido que tahn()

3.3 Training on multiple GPUs

Due to the large size of the dataset (1.2 million training examples), AlexNet was trained across two GPUs. This parallelization scheme split the network, with each GPU handling half of the neurons. The GPUs communicated only in certain layers, facilitating efficient training without significant memory bottlenecks.

3.4 Local Response Normalization

ReLUs have the property that they do not require input normalization to prevent them from saturating. The LRN technique is applied after ReLU activation in the first and second conventional layers, constrains the neuron's responses and prevents them from becoming too big, in order to normalize the responses and enhance the model's generalization ability. The normalization across channels contributes to this result, as any neuron on the feature map has its response computed divided by the sum of the squared responses of neurons in a nearby region across all channels.

3.5 Overlapping pooling

Pooling layers in CNNs summarize the outputs of neighboring groups of neurons in the same kernel map.

Max-pooling layers are present, follow the convolutional layers to reduce spatial dimensions and create a form of translation invariance. They have 3x3 size and stride of 2.

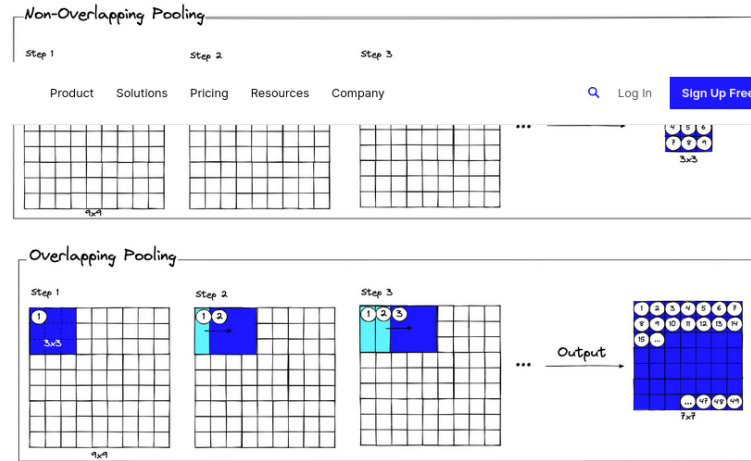


Fig. 2

After convolutional and max-pooling layers, there are three fully connected layers with 4096 neurons each. They act like classifiers and learn high-level features from the extracted features.

3.6 Fully Connected Layers

After the convolutional and max-pooling layers, AlexNet includes three fully connected layers, each with 4096 neurons. These layers act as classifiers, learning high-level features from the extracted features. The final layer is a 1000-node softmax, producing a probability distribution over 1000 classes.

4 Main concern- reducing overfitting

Because one neural network architecture has 60 million parameters it can be difficult to learn them all without overfitting. There are ways to combat it.

4.1 Data Augmentation

One easy way to reduce overfitting on image data is to artificially enlarge the dataset using label-preserving transformations. There are two forms of data augmentation that are employed in AlexNet, both allowing transformed images to be produced from the original image with very little computation. The transformed images do not need to be stored on disk and are generated in Python code on the CPU while the GPU is training on the previous batch of images. The benefit of data augmentation schemes is that they are computationally free.

First we have generating image transitions and horizontal reflections. In AlexNet, they extract random 224×224 patches and their horizontal reflections from the 256×256 images and train the network on those extracted patches. It results in a bigger training set, but the resulting training examples are highly interdependent. Without this, the network suffers from massive overfitting. At test time, the network makes a prediction

by extracting five 224×224 patches, as well as their horizontal reflections (so ten patches in all), and averaging the predictions made by the network's softmax layer on the ten patches.

Second, we have the altering of the intensities of the RGB channels in training images. They perform PCA on the set of RGB pixel values throughout the ImageNet training set. To each training image, they add multiples of the found principle components, with magnitudes proportional to the corresponding eigenvalues times a random variable drawn from a Gaussian with mean zero and standard deviation 0.1. So any RGB image pixel $xy = [IR_{xy}, IG_{xy}, IB_{xy}]^T$. Each i is drawn only once for all the pixels of a particular training image until that image is used for training again, at which point it is re-drawn. This scheme approximately captures an important property of natural images, namely, that object identity is invariant to changes in the intensity and color of the illumination. This scheme reduces the top-1 error rate by over 1

4.2 Dropout

Combining predictions from multiple models is a highly effective method for reducing test errors [1, 3]. However, this approach can be prohibitively expensive for large neural networks that require several days to train. An efficient alternative is a model combination technique known as "dropout" [10], which only doubles the training cost. Dropout involves randomly setting the output of each hidden neuron to zero with a probability of 0.5. Neurons that are "dropped out" do not contribute to the forward pass and are excluded from back-propagation. Consequently, the neural network uses a different architecture for each input, with all these architectures sharing the same weights. This approach mitigates complex co-adaptations of neurons, compelling each neuron to learn more robust features that are effective with various random subsets of other neurons. During testing, all neurons are utilized, but their outputs are scaled by 0.5, approximating the geometric mean of the predictive distributions from the numerous dropout networks.

5 Details

5.1 Stochastic gradient Descent

AlexNet utilizes stochastic gradient descent as its optimization algorithm. Its commonly used to train deep learning models, including convolutional neural networks. Its use in AlexNet helps in updating the models weights iteratively based on small batches of data, which aids in faster convergence and offer better generalization compared to using the entire dataset for each update.

In AlexNet, specific variations and enhancements to standard SGD were also implemented, such as:

- Momentum: This technique helps accelerate gradients vectors in the right directions, thus leading to faster converging. It reduces the oscillations and smooths the updates.

- Learning Rate Scheduling: The learning rate is adjusted during the training process. AlexNet initially starts with a higher learning rate, which is then gradually decreased over time to allow for more fine-tuned learning.

These enhancements to the basic SGD algorithm contribute to the effective training of AlexNet, enabling it to achieve high performance on challenging image recognition tasks.

6 Results

6.1 Performance on the on ILSVRC-2010

The results on the ILSVRC-2010 competition dataset are presented in Table 1. The network achieves top-1 and top-5 test set error rates of 37.5% and 17.0%, respectively. During the ILSVRC-2010 competition, the best performance recorded was 47.1% and 28.2%, achieved by averaging predictions from six sparse-coding models trained on different features [2]. Subsequently, the best published results were 45.7% and 25.7%, obtained by averaging predictions from two classifiers trained on Fisher Vectors (FVs) computed from two types of densely-sampled features [24].

Table 1 Caption text

Model	Top-1	Top-5
Sparse coding[2]	47.1%	28.2%
SIFT + FVs[24]	45.7%	25.7% ¹
CNN	37.5%	17.0% ²

Source: AlexNet paper

6.2 Performance on ILSVRC-2012

In the ILSVRC-2012 competition, AlexNet achieved a top-5 error rate of 18.2% on the validation set. The network's performance was further enhanced by averaging the predictions of five similar CNNs, which reduced the top-5 error rate to 16.4%. When trained on the entire ImageNet Fall 2011 release (15 million images, 22,000 categories) and fine-tuned on ILSVRC-2012, a single CNN with an additional sixth convolutional layer achieved a top-5 error rate of 16.6%. Averaging the predictions of these two pre-trained CNNs with the five previously trained CNNs resulted in an even lower top-5 error rate of 15.3

Table 2 Performance on the ILSVRC-2012

Model	Top-1(val)	Top-5(val)	Top-5(test)
SIFT+FVs	—	—	26.2%
1 CNN	40.7%	18.2%	—
5 CNNs	38.1%	16.4%	16.4%
1 CNN*	39%	16.6%	—
7 CNNs*	36.7%	15.4%	15.3%

Models with an asterisk (*) were "pre-trained" to classify the entire ImageNet 2011 Fall release.

6.3 Performance on ImageNet Fall 2009

AlexNet’s performance was also evaluated on the Fall 2009 version of ImageNet, which contains 10,184 categories and 8.9 million images. The network, with an additional sixth convolutional layer, achieved top-1 and top-5 error rates of 67.4% and 40.9%, respectively. The best published results on this dataset prior to AlexNet were 78.1% and 60.9% .

6.4 Qualitative Evaluations

AlexNet’s ability to learn useful features was demonstrated qualitatively through visualizations of the convolutional kernels learned by the network’s two data-connected layers. The kernels displayed a variety of frequency- and orientation-selective patterns, as well as various colored blobs. Interestingly, due to the network’s design, the kernels on one GPU were largely color-agnostic, while those on the other GPU were color-specific. This specialization occurred consistently across different runs and random weight initializations .

In Figure 3 of the paper, the top-5 predictions for eight test images are shown. Even off-center objects, such as a mite, were recognized accurately. Most of the top-5 labels appeared reasonable, with plausible alternatives provided for each image. For example, only other types of cats were considered plausible labels for a leopard. Some images, like those of a grille or a cherry, had genuine ambiguity regarding their focus, demonstrating the model’s nuanced understanding of visual content .

Another qualitative evaluation involved computing the Euclidean distance between feature activation vectors in the last hidden layer (4096-dimensional) to measure the similarity between images. This method highlighted the network’s ability to capture semantic similarities rather than just pixel-level similarities. For instance, retrieved images of dogs and elephants appeared in various poses, indicating that the network learned to recognize objects despite variations in pose and appearance .

6.5 Training Efficiency

Training AlexNet was computationally intensive, requiring roughly 90 cycles through the 1.2 million images in the training set, which took five to six days on two NVIDIA GTX 580 3GB GPUs. The learning rate was reduced three times prior to the termination of training to ensure convergence and stability of the model .

7 Conclusion

AlexNet’s groundbreaking achievements in the ILSVRC-2012 competition heralded a new era for deep learning and computer vision. By leveraging innovative architectural choices, training strategies, and techniques to combat overfitting, AlexNet set new benchmarks for performance and practicality. Its legacy endures in the ongoing advancements in deep learning, inspiring researchers and practitioners to push the boundaries of what is possible .

8 Cross referencing

8.1 Details on reference citations

All cited bib entries are printed at the end of this article:[1] [2].

References

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